Linear Algebra & Differential Equations

Date

Assignment # 4

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SECTION : J (CE)

Q1. $y = c_1 e^{x} + c_2 e^{-x}$

$\Rightarrow y(0) = c_1 e^{0} + c_2 e^{0} = c_1 + c_2 = 0$

$\Rightarrow y' = c_1 e^{x} - c_2 e^{-x}$

$\Rightarrow y'(0) = c_1 e^{0} - c_2 e^{0} = 1$

$\Rightarrow c_1 - \cancel{c_2} = 1 \quad - (1)$; $1/2 + c_2 = 0$

$\Rightarrow + \; c_1 + \cancel{c_2} = 0 \quad - (2)$; $c_2 = -1/2$

$2c_1 = 1$

$c_1 = 1/2$

Substituting c_1 and c_2 back into the general solution :

$y = \frac{1}{2} e^{x} - \frac{1}{2} e^{-x}$

Q2. $y = c_1 e^{4x} + c_2 e^{-x}$

$\Rightarrow y(0) = c_1 e^{4(0)} + c_2 e^{0} = 1$

$\Rightarrow c_1 + c_2 = 1$

$\Rightarrow y' = 4c_1 e^{4x} - c_2 e^{-x}$

$\Rightarrow y'(0) = 4c_1 e^{0} - c_2 e^{0} = 2$

$\Rightarrow 4c_1 - \cancel{c_2} = 2 \quad - (1)$; $3/5 + c_2 = 1$

$\Rightarrow + \; c_1 + \cancel{c_2} = 1 \quad - (2)$; $c_2 = 1 - 3/5 = 2/5$

$5c_1 = 3$

$c_1 = 3/5$

Substituting c_1 and c_2 back into the general solution :

$y = 3/5 e^{4x} + 2/5 e^{-x}$

Q3. $y = c_1 x + c_2 x \ln x$

$\Rightarrow y(1) = c_1(1) + c_2(1)\ln 1 = 3$

$\Rightarrow c_1 = 3$

$\Rightarrow y' = c_1 + c_2\left[(x \ln x)\, d/dx\right]$

$y' = c_1 + c_2\left[x(1/x) + \ln x(1)\right]$

$y' = c_1 + c_2\left[\ln x + 1\right]$

$y' = c_1 + c_2[\ln x + 1]$

$\Rightarrow \cancel{y'(1) = c_1 + c_2 \ln(1) + 1}\; y'(1) = c_1 + c_2[\ln 1 + 1] = -1$

$\Rightarrow c_1 + c_2(1) = -1$

$\Rightarrow c_1 + c_2 = -1$

$\Rightarrow 3 + c_2 = -1$

$c_2 = -1 - 3 = -4$

$\therefore c_1 = 3$

$c_2 = -4$

Substituting c_1 and c_2 into the general solution ;

$\Rightarrow y = 3x - 4x \ln x$

Q4. $y = c_1 + c_2 \cos x + c_3 \sin x$

$\Rightarrow y(\pi) = c_1 + c_2 \cos\pi + c_3 \sin\pi = 0$

$\Rightarrow c_1 + c_2(-1) + c_3(0) = 0$

$\Rightarrow c_1 - c_2 = 0$

$\Rightarrow y' = 0 - c_2 \sin x + c_3 \cos x$

$\Rightarrow y'(\pi) = -c_2 \sin\pi + c_3 \cos\pi = \cancel{1}\, 2$

$\Rightarrow -c_2(0) + c_3(-1) = \cancel{1}\, 2$

$\Rightarrow -c_3 = \cancel{1}\, 2$

$\Rightarrow c_3 = \cancel{1} -2$

$\Rightarrow y'' = -c_2 \cos x - c_3 \sin x$

$\Rightarrow y''(\pi) = -c_2 \cos\pi - c_3 \sin\pi = -1$

$\Rightarrow -c_2(-1) - c_3(0) = -1$

$\Rightarrow c_2 = -1$

$\therefore c_1 + 1 = 0$

$c_1 = -1$

Substituting c_1, c_2, and c_3 into the general solution:

$\Rightarrow y = -1 - \cos x - 2\sin x$

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15. $f_1(x) = x$
$f_2(x) = x^2$
$f_3(x) = 4x - 3x^2$

$$\begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ f_1'(x) & f_2'(x) & f_3'(x) \\ f_1''(x) & f_2''(x) & f_3''(x) \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} x & x^2 & 4x - 3x^2 \\ 1 & 2x & 4 - 6x \\ 0 & 2 & -6 \end{vmatrix} = x\left(-12x - 2(4-6x)\right) - 1\left[-6x^2 - 2(4x - 3x^2)\right] + 0$$

$\Rightarrow x(-12x - 8 + 12x) - 1[-6x^2 - 8x \mp 6x^2]$

$\Rightarrow -12x^2 - 8x + 12x^2 + 6x^2 + 8x \mp 6x^2$

$\Rightarrow -12x^2 + 12x^2 - 8x + 8x + 6x^2 - 6x^2 = 0$

Since $W = 0$, $\therefore$ the set of functions are linearly dependent.

16. $f_1(x) = 0$
$f_2(x) = x$
$f_3(x) = e^x$

Linearly dependent because, any set containing the zero function is automatically linearly dependent because 0 can be written as $(0 \times f_2) + (0 \times f_3) = 0$.

17. $f_1(x) = 5$
$f_2(x) = \cos^2 x$
$f_3(x) = \sin^2 x$

$$\begin{vmatrix} 5 & \cos^2 x & \sin^2 x \\ 0 & -\sin(2x) & 2\sin(2x) \\ 0 & -2\cos(2x) & 2\cos(2x) \end{vmatrix}$$

~~$\Rightarrow 5[\sin^2 x + \cos^4 x]] + (0)(0)$~~

~~$\Rightarrow 5 [\sin^2 x + \sin^2 x]$~~

$\Rightarrow 5\left[(-\sin(2x))(2\cos(2x)) + (2\cos(2x))(2\sin(2x))\right]$

$\Rightarrow 5\left[-2\sin(2x)\cos(2x) + 2\sin(2x)\cos(2x)\right] = 0$

For arbitary functions, this is inconclusive, but it can be verified;

$\Rightarrow f_1(x) = 5 = 5(\cos^2 x + \sin^2 x) = 5f_2(x) + 5f_3(x)$.

Since $f_1(x)$ is a linear combination of $f_2(x)$ and $f_3(x)$, the set of functions are Linearly dependent.

Q18. $f_1(x) = \cos 2x$

$f_2(x) = 1$

$f_3(x) = \cos^2 x$

$\cos 2x = 2\cos^2 x - 1$

$\therefore f_1(x) = \cos 2x$

$f_1(x) = 2\cos^2 x - 1$

$f_1(x) = 2 f_3(x) - f_2(x)$

Since $f_1(x)$ is a linear combination of $f_2(x)$ and $f_3(x)$, the set of functions are linearly dependent.

Q19. $f_1(x) = x$

$f_2(x) = x - 1$

$f_3(x) = x + 3$

$$\begin{vmatrix} x & x-1 & x+3 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{vmatrix}$$

PROPERTY of a DETERMINANT : If a determinant has any two rows (or columns) identical, then its value is zero.

$$\therefore \begin{vmatrix} x & x-1 & x+3 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

For arbitary functions, the wronskian determinant being zero leads to inconclusiveness.

Since there is no function that can be expressed as a linear combination of the other two; the set of functions is linearly independent.

Q20. $f_1(x) = 2 + x$

$f_2(x) = 2 + |x|$

$$\begin{vmatrix} 2+x & 2+|x| \\ 1 & \frac{x}{|x|} \end{vmatrix}$$

$\Rightarrow (2+x)\left(\frac{x}{|x|}\right) - (2+|x|) = \frac{(2+x)(x)}{|x|} - (2+|x|)$

NOT POSSIBLE !

The function $|x|$ is not differentiable at $x = 0$, but linear dependence can still be checked.

Suppose $c_1(2+x) + c_2(2+|x|) = 0$ for all x;

$\Rightarrow$ For $x > 0$, $c_1(2+x) + c_2(2+x) = 0$

$(c_1 + c_2)(2+x) = 0$; $c_1 = -c_2$

$\Rightarrow$ For $x < 0$, $c_1(2+x) + c_2(2-x) = 0$; if $c_1 = -c_2$

$\therefore -c_2(2+x) + c_2(2-x)$

$= -2c_2 - c_2 x + 2c_2 - c_2 x$

This must hold for all $x < 0$, so $c_2 = 0$, and thus $c_1 = 0$.

$\therefore$ linearly independent.

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21. $f_1(x) = 1 + x$

$f_2(x) = x$

$f_3(x) = x^2$

The functions are linearly independent because they are a basis for the space of quadratic polynomials.
No non-trivial combination yields zero for all x.

1. $f_1(x) = e^x$; $\sinh x = \frac{e^x - e^{-x}}{2}$

$f_2(x) = e^{-x}$

$f_3(x) = \sinh x$

$\therefore f_3(x) = \sinh x = \frac{e^x - e^{-x}}{2} = \frac{e^x}{2} - \frac{e^{-x}}{2}$

$\Rightarrow f_3(x) = \frac{1}{2} f_1(x) - \frac{1}{2} f_2(x)$

Since $f_3(x)$ is a linear combination of $f_1(x)$ and $f_2(x)$, the set is linearly dependent.

1. $y'' - 7y' + 10y = 24e^x$

$\Rightarrow$ The solution will be of the form;

$$y = y_c + y_p$$

For the complementary solution;

The characteristic eq. of the complementary (homogenous) part;

$\Rightarrow m^2 - 7m + 10 = 0$ } from : $y'' - 7y' + 10y$

$\Rightarrow m^2 - 5m - 2m + 10 = 0$

$\Rightarrow m(m-5) - 2(m-5) = 0$

$\Rightarrow (m-2)(m-5) = 0$

$m_1 = 2$, $m_2 = 5$

$\therefore$ Complementary sol. $\Rightarrow y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

$\Rightarrow y_c = c_1 e^{2x} + c_2 e^{5x}$

For the particular solution;

Since the non-homogenous part of the eq. in question is exponential ;

$\Rightarrow$ $y_p = Ae^x$; $y_p'' = Ae^x$

$y'p = Ae^x$

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$\therefore (Ae^x)'' - 7(Ae^x)' + 10(Ae^x) = 24e^x$

$\Rightarrow Ae^x - 7Ae^x + 10Ae^x = 4Ae^x = 24e^x$

$\therefore A = 6$

$\therefore y_P = 6e^x$

$\therefore y = y_c + y_P$

$y = c_1e^{2x} + c_2e^{5x} + 6e^x$ Q.E.D

Q32. $y'' + y = \sec x$

For the complementary solution ;

The characteristic eq. of the complementary (homogenous) part ;

$\Rightarrow m^2 + 1 = 0$

$\Rightarrow m^2 = -1$

$\Rightarrow m = \pm i = 0 \pm i$; $\alpha = 0$; $b = 1$

$\therefore$ Since the roots are complex conjugate ;

$\Rightarrow y_c = e^0(c_1\cos(x) + c_2\sin(x))$

$y_c = c_1\cos x + c_2\sin x$

$\Rightarrow y_1 = \cos x$

$y_2 = \sin x$

Since the nonhomogenous term $g(x) = \sec x$ is NOT a polynomial, exponential, sine, cosine, or a combination of these ;

Applying the variation of parameter ;

$f(x) = \sec x$

$$y_p = u_1y_1 + u_2y_2 \qquad W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}$$

$$u_1' = \frac{W_1}{W}, \quad u_2' = \frac{W_2}{W} \qquad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W_1 = \begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix}$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix}$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$W = \cos^2 x + \sin^2 x$; $W_1 = 0 - \sin x \left(\frac{1}{\cos x}\right) = -\tan x$

$W = 1$

$W_2 = \cos x \left(\frac{1}{\cos x}\right) - 0 = 1$

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$\Rightarrow \int u_1' dx = \int -\tan x \, dx$

$\Rightarrow u_1 = \int -\frac{\sin x}{\cos x} dx = \ln|\cos x| = u_1$

$\Rightarrow \int u_2' dx = \int 1 \, dx = x = u_2$

$\therefore y_p = \ln|\cos x| (\cos x) + x \sin x$

$\Rightarrow y = c_1 \cos x + c_2 \sin x + \ln(\cos x)(\cos x) + x \sin x$ $8(\cdot 1)$

• $y'' - 4y' + 4y = 2e^{2x} + 4x - 12$

For the complementary solution :

$\Rightarrow m^2 - 4m + 4 = 0$

$\Rightarrow m^2 - 2m - 2m + 4 = 0$; $m(m-2) - 2(m-2) = 0$

$\Rightarrow (m-2)(m-2) = 0$; $m = 2, 2$

$\therefore y_c = c_1 e^{2x} + c_2 x e^{2x}$

For the particular solution ;

The nonhomogenous term $g(x) = 2e^{2x} + 4x - 12$, contains both exponential and polynomial terms ;

For the exponential term e^{kx} :

$\Rightarrow$ If e^{kx} is NOT part of the complementary solution ;

$\Rightarrow y_{p_1} = Ae^{kx}$

$\Rightarrow$ If e^{kx} is part of the complementary solution ;

$\Rightarrow y_{p_1} = Ax^{\alpha} e^{kx}$

where α = multiplicity of the root in the characteristic eq.

For the polynomial term $P(x)$:

$\Rightarrow$ If $P(x) = 4x - 12$ (degree 1) ;

$\Rightarrow y_{p_2} = Bx + C$

$\Rightarrow$ If $P(x) = x^2 + 3x$ (degree 2) ;

$\Rightarrow y_{p_2} = Dx^2 + Ex + F$

• $y_p = y_{p_1} + y_{p_2}$

$\therefore$ for $2e^{2x}$; $y_{p_1} = Ax^2 e^{2x}$

For $4x - 12$; $y_{p_2} = Bx + C$

$\Rightarrow y_p = Ax^2 e^{2x} + Bx + C$

$y'_p = A\left[2xe^{2x} + x^2 2e^{2x}\right]$

$y'_p = 2Axe^{2x} + 2Ax^2 e^{2x} + B$

Rainbow

Teacher's Sign.

$y''_p = 2A[x2e^{2x} + e^{2x}] + 2A[2xe^{2x} + x^2 2e^{2x}]$

$y''_p = 4Axe^{2x} + 2Ae^{2x} + 4Axe^{2x} + 4Ax^2e^{2x}$

$y''_p = 8Axe^{2x} + 2Ae^{2x} + 4Ax^2e^{2x}$

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$\Rightarrow$ Substitute into $y'' - 4y' + 4y = 2e^{2x} + 4x - 12$

$\Rightarrow [8Axe^{2x} + 2Ae^{2x} + 4Ax^2e^{2x}] - 4[2Axe^{2x} + 2Ax^2e^{2x} + B] + 4[Ax^2e^{2x} + Bx + C]$

$\Rightarrow 2Ae^{2x} - 4B + 4Bx + 4C = 2e^{2x} + 4x - 12$

$\Rightarrow 2Ae^{2x} = 2e^{2x}$; $4Bx - 4B + 4C = 4x - 12$

$\Rightarrow A = 1$; $4Bx - 4(B-C) = 4x - 12$

$\Rightarrow 4Bx = 4x$; $-4(1-C) = -12$

$B = 1$; $\Rightarrow C = -2$

$\therefore y_p = x^2e^{2x} + x - 2$

$\Rightarrow \therefore y = c_1e^{2x} + c_2xe^{2x} + x^2e^{2x} + x - 2$ Q.E.D

General Solution of Second-Order Differential Equation

Q1. $4y'' + y' = 0$

$\Rightarrow 4m^2 + m = 0$

$\Rightarrow m(4m+1) = 0$

$\Rightarrow m = 0$, $4m + 1 = 0$

$m = 0$, $m = -1/4$

$\therefore y = c_1e^0 + c_2e^{-1/4}$

$y = c_1 + c_2e^{-1/4x}$

Q2. $y'' - 36y = 0$

$\Rightarrow m^2 - 36 = 0$

$\Rightarrow (m+6)(m-6) = 0$

$\Rightarrow m = -6$, $m = 6$

$\therefore y = c_1e^{6x} + c_2e^{-6x}$

Q3. $y'' - y' - 6y = 0$

$m^2 - m - 6 = 0$

$\Rightarrow m^2 - 3m + 2m - 6 = 0$

$\Rightarrow m(m-3) + 2(m-3) = 0$

$\Rightarrow (m+2)(m-3) = 0$

$\Rightarrow m = -2$, $m = 3$

$\therefore y = c_1e^{-2x} + c_2e^{3x}$

Q4. $y'' - 3y' + 2y = 0$

$\Rightarrow m^2 - 3m + 2 = 0$

$\Rightarrow m^2 - 2m - m + 2 = 0$

$\Rightarrow m(m-2) - 1(m-2) = 0$

$(m-1)(m-2) = 0$

$\therefore y = c_1e^{x} + c_2e^{2x}$

5. $y'' + 8y' + 16y = 0$

$\Rightarrow m^2 + 8m + 16 = 0$

$\Rightarrow m^2 + 4m + 4m + 16 = 0$

$\Rightarrow m(m+4) + 4(m+4) = 0$

$(m+4)(m+4) = 0$

$\Rightarrow m = -4$

$\therefore y = c_1 e^{-4x} + c_2 x e^{-4x}$

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$y'' - 10y' + 25y = 0$

$\Rightarrow m^2 - 10m + 25 = 0$

$\Rightarrow m^2 - 5m - 5m + 25 = 0$

$\Rightarrow m(m-5) - 5(m-5) = 0$

$\Rightarrow (m-5)(m-5) = 0$

$m = 5$

$\therefore y = c_1 e^{5x} + c_2 x e^{5x}$

$12y'' - 5y' - 2y = 0$

$\Rightarrow 12m^2 - 5m - 2 = 0$

$\Rightarrow 12m^2 - 8m + 3m - 2 = 0$

$\Rightarrow 4m(3m-2) + 1(3m-2) = 0$

$\Rightarrow (4m+1)(3m-2) = 0$

$m = -1/4, \quad m = 2/3$

$\therefore y = c_1 e^{-1/4x} + c_2 e^{2/3x}$

$y'' + 4y' - y = 0$

$m^2 + 4m - 1 = 0$

$\Rightarrow m \Rightarrow \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \dfrac{-4 \pm \sqrt{16 - 4(1)(-1)}}{2} = \dfrac{-4 \pm \sqrt{20}}{2}$

$m \Rightarrow \dfrac{-4 \pm \sqrt{20}}{2} = -2 + \sqrt{5}, \; -2 - \sqrt{5}$

$\therefore y = c_1 e^{(-2+\sqrt{5})x} + c_2 e^{(-2-\sqrt{5})x}$

VARIATION of PARAMETER

$y'' + y = \sec x$

For the complementary sol. :

$m^2 + 1 = 0$

$\Rightarrow m^2 = -1$

$\Rightarrow m = \pm i = 0 \pm i \; ; \; a = 0 \; , \; b = 1$

$\therefore y_c = e^0 (c_1 \cos x + c_2 \sin x)$

$y_c = c_1 \cos x + c_2 \sin x$

$y_c = c_1 \cos x + c_2 \sin x$

$y_1 = \cos x$

$y_2 = \sin x$

$y_p = u_1 y_1 + u_2 y_2$ $\quad W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}$ $\quad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

$u_1' = \frac{W_1}{W}, \quad u_2' = \frac{W_2}{W}$ $\quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$

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$$W_1 = \begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix} = 0 - \sec x(\sin x) = -\sin x\left(\frac{1}{\cos x}\right) = \ldots$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix} = \cos x\left(\frac{1}{\cos x}\right) - 0 = 1$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$\int u_1' dx = \int -\tan x \, dx$$

$$u_1 = \int \frac{-\sin x}{\cos x} dx = \ln|\cos x| = u_1$$

$$\int u_2' dx = \int 1 \, dx = x = u_2$$

$\therefore \; y_p = \ln|\cos x| \cos x + x \sin x$

$\Rightarrow \; y = c_1 \cos x + c_2 \sin x + \ln(\cos x)\cos x + x \sin x$

Q2. $y'' + y = \tan x$

For the complementary sol. ;

$\Rightarrow m^2 + 1 = 0$ $\quad \therefore \; y_c = c_1 \cos x + c_2 \sin x$

$\Rightarrow m = \pm i$

$\Rightarrow y_c = c_1 \cos x + c_2 \sin x$

$\Rightarrow \therefore \; y_1 = \cos x$

$y_2 = \sin x$

$$W_1 = \begin{vmatrix} 0 & \sin x \\ \tan x & \cos x \end{vmatrix} = 0 - \left(\frac{\sin x}{\cos x}\right)(\sin x) = -\frac{\sin^2 x}{\cos x}$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \tan x \end{vmatrix} = \cos x\left(\frac{\sin x}{\cos x}\right) - 0 = \sin x$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$\Rightarrow \int u_1' dx = \int \frac{-\sin^2 x}{\cos x} dx = \sin x - \ln(\tan x + \sec x) = u_1$$

$$\Rightarrow \int u_2' dx = \int \sin x \, dx = -\cos x = u_2$$

$\therefore \; y_p = [\sin x - \ln(\tan x + \sec x)]\cos x + [-\cos x]\sin x$

$\Rightarrow \; y = c_1 \cos x + c_2 \sin x + \cos x[\sin x - \ln(\tan x + \sec x)] - \cos x \sin x$

Q3. $y'' + y = \sin x$

$m^2 + 1 = 0$ $\Rightarrow$ $y_1 = \cos x$

$y_2 = \sin x$

$\therefore$ $y_c = c_1 \cos x + c_2 \sin x$

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$W_1 = \begin{vmatrix} 0 & \sin x \\ \sin x & \cos x \end{vmatrix} = 0 - \sin^2 x = -\sin^2 x$

$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sin x \end{vmatrix} = \cos x \sin x - 0 = \cos x \sin x$

$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$

$\int u_1' dx = \int -\sin^2 x \, dx = \frac{\sin(2x) - 2x}{4} = u_1$

$\int u_2' dx = \int \cos x \sin x \, dx = \frac{\sin^2(x)}{2} \cancel{+} = u_2$

$\Rightarrow$ $y_P = \left(\frac{\sin(2x) - 2x}{4}\right)(\cos x) + \left(\frac{\sin^2(x)}{2}\right)(\sin x)$

$\Rightarrow$ $y = c_1 \cos x + c_2 \sin x + \left(\frac{\sin(2x) - 2x}{4}\right)(\cos x) + \left(\frac{\sin^2(x)}{2}\right)(\sin x)$

$y'' + y = \sec\theta \tan\theta$ $\Rightarrow$ $y_1 = \cos x$

$m^2 + 1 = 0$ $\quad$ $y_2 = \sin x$

$\therefore$ $y_c = c_1 \cos x + c_2 \sin x$

$W_1 = \begin{vmatrix} 0 & \sin x \\ \sec x \tan x & \cos x \end{vmatrix} = -\sin x \left(\frac{1}{\cos x}\right)\left(\frac{\sin x}{\cos x}\right) = \frac{-\sin^2 x}{\cos^2 x}$

$W_1 = -\tan^2 x$

$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \tan x \end{vmatrix} = \cos x \left(\frac{1}{\cos x}\right)\left(\frac{\sin x}{\cos x}\right) = \tan x$

$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$

$\int u_1' dx = \int -\tan^2 x \, dx = x - \tan(x) = u_1$

$\int u_2' dx = \int \tan x \, dx = -\ln|\cos x|$

$\Rightarrow$ $y_P = (x - \tan x)(\cos x) + (-\ln(\cos x))(\sin x)$

$\Rightarrow$ ~~$y = $~~

$\Rightarrow$ $y = c_1 \cos x + c_2 \sin x + \cos x(x - \tan x) - \sin x(\ln(\cos x))$

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Q5. $y'' + y = \cos^2 x$ $\Rightarrow y_1 = \cos x$

$m^2 + 1 = 0$ $y_2 = \sin x$

$\therefore y_c = c_1 \cos x + c_2 \sin x$

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$$W_1 = \begin{vmatrix} 0 & \sin x \\ \cos^2 x & \cos x \end{vmatrix} = 0 - \sin x (\cos^2 x) = -\sin x (\cos^2 x)$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \cos^2 x \end{vmatrix} = (\cos^2 x)(\cos x) - 0 = \cos^3 x$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$\int u_1' dx = \int -\sin x (\cos^2 x) dx = \frac{\cos^3(x)}{3} = u_1$$

$$\int u_2' dx = \int \cos^3 x \, dx = \sin(x) - \frac{\sin^3(x)}{3} = u_2$$

$$\Rightarrow y_P = \left(\frac{\cos^3(x)}{3}\right)(\cos x) + \left[\sin(x) - \frac{\sin^3(x)}{3}\right](\sin x)$$

$$\Rightarrow y = c_1 \cos x + c_2 \sin x + \left(\frac{\cos^3(x)}{3}\right)(\cos x) + \left[\sin(x) - \frac{\sin^3(x)}{3}\right](s$$

Q6. $y'' + y = \sec^2 x$ $\Rightarrow y_1 = \cos x$

$m^2 + 1 = 0$ $y_2 = \sin x$

$\therefore y_c = c_1 \cos x + c_2 \sin x$

$$W_1 = \begin{vmatrix} 0 & \sin x \\ \sec^2 x & \cos x \end{vmatrix} = -\sin x \left(\frac{1}{\cos^2 x}\right) = -\frac{\sin x}{\cos x}\left(\frac{1}{\cos x}\right)$$

$W_1 = -\tan x \sec x$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sec^2 x \end{vmatrix} = \cos x \left(\frac{1}{\cos^2 x}\right) - 0 = \sec x$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$\int u_1' dx = \int -\tan x \sec x \, dx = -\sec x = u_1$$

$$\int u_2' dx = \int \sec x \, dx = \int \frac{\sec x}{1} \times \frac{(\sec x + \tan x)}{\sec x + \tan x} dx$$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx \quad ; \quad u = \sec x + \tan x$$

$du = \sec x \tan x + \sec^2 x \, dx$

Rainbow

$\therefore \int \frac{du}{u} = \int \frac{1}{u}\, du = \ln u$ $\qquad \sec x = \left(\frac{1}{\cos x}\right)(\cos x)$

$\Rightarrow \ln|\sec x + \tan x| = u_2$

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$\therefore \Rightarrow y_P = -\sec x(\cos x) + (\ln|\sec x + \tan x|)(\sin x)$

$\Rightarrow y = c_1 \cos x + c_2 \sin x - 1 + \sin x\,[\ln|\sec x + \tan x|]$

$y'' - y = \cosh x$

$\Rightarrow m^2 - 1 = 0$

$\Rightarrow m^2 = 1$

$m = \pm 1$

$\therefore y_c = c_1 e^x + c_2 e^{-x}$

$\Rightarrow y_1 = e^x$

$\Rightarrow y_2 = e^{-x}$

$W_1 = \begin{vmatrix} 0 & e^{-x} \\ \cosh x & -e^{-x} \end{vmatrix}$

$W_2 = \begin{vmatrix} e^x & 0 \\ e^x & \cosh x \end{vmatrix}$

$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix}$

$\Rightarrow W_1 = -\cosh x\,(e^{-x}) = -\left(\frac{e^x + e^{-x}}{2}\right)(e^{-x})$

$W_1 = \frac{-e^x e^{-x} + -e^{-x} e^{-x}}{2} = \frac{-e^0 - e^{-2x}}{2} = \frac{-1 - e^{-2x}}{2} = \frac{-1(1+e^{-2x})}{2}$

$W_2 = e^x\left(\frac{e^x + e^{-x}}{2}\right) = \frac{e^{2x} + 1}{2}$

$W = -1 - 1 = -2$

$\int u_1'\,dx = \int \frac{-1(1+e^{-2x})}{2} \times \frac{1}{-2} = \frac{1}{4}\int 1 + e^{-2x}\,dx = \frac{1}{4}\left[x - 2e^{-2x}\right]$

$\Rightarrow u_1 = 1/4\,(x - 2e^{-2x})$

$\int u_2'\,dx = \int \frac{e^{2x}+1}{2} \times \frac{1}{-2} = \frac{-1}{4}\int e^{2x} + 1\,dx = \frac{-1}{4}\left[2e^{2x} + x\right]$

$\Rightarrow u_2 = -1/4\,(2e^{2x} + x)$

$\Rightarrow y_P = \left[1/4\,(x - 2e^{-2x})\right](e^x) + \left[-1/4\,(2e^{2x} + x)\right](e^{-x})$

$y_P = \frac{1}{4}(x - 2e^{-2x})(e^x) - \frac{1}{4}(2e^{2x} + x)(e^{-x})$

$y_P = \frac{xe^x - 2e^{-2x+x}}{4} - \left(\frac{2e^{2x-x} + xe^{-x}}{4}\right) = \frac{xe^x - 2e^{-x}}{4} - \left(\frac{2e^x + xe^{-x}}{4}\right)$

$\Rightarrow y = c_1 e^x + c_2 e^{-x} + \frac{1}{4}\left[(xe^x - 2e^{-x}) - (2e^x + xe^{-x})\right]$

Rainbow

1 Q8. $y'' - y = \sinh 2x$ $\Rightarrow$ $y_1 = e^{x}$, $y_2 = e^{-x}$

$m^2 - 1 = 0$

$\therefore y_c = C_1 e^{x} + C_2 e^{-x}$

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$$W_1 = \begin{vmatrix} 0 & e^{-x} \\ \sinh 2x & -e^{-x} \end{vmatrix} = -\left(\frac{e^{2x} - e^{-2x}}{2}\right)(e^{-x})$$

$$= \frac{-e^{2x}e^{-x} + e^{-2x}e^{-x}}{2} = \frac{-e^{x} + e^{-3x}}{2}$$

$\sinh(x) = \frac{e^{x} - e^{-x}}{2}$

$\Rightarrow \sinh(2x) = \frac{e^{2x} - e^{-2x}}{2}$

$$\Rightarrow W_2 = \begin{vmatrix} e^{x} & 0 \\ e^{x} & \sinh 2x \end{vmatrix} = e^{x}\left(\frac{e^{2x} - e^{-2x}}{2}\right) = \frac{e^{x}e^{2x} - e^{x}e^{-2x}}{2}$$

$$W_2 = \frac{e^{3x} - e^{-x}}{2}$$

$$\Rightarrow W = \begin{vmatrix} e^{x} & e^{-x} \\ e^{x} & -e^{-x} \end{vmatrix} = -2$$

$$\int u_1' dx = \int \frac{-e^{x} + e^{-3x}}{2} \times \frac{1}{-2} = \frac{-1}{4} \int -e^{x} + e^{-3x} dx = \frac{-1}{4}\left[-3e^{-3x} - e^{x}\right]$$

$$\Rightarrow u_1 = \frac{-1}{4}\left(-3e^{-3x} - e^{x}\right) = \frac{-1}{4}(-1)\left(3e^{-3x} + e^{x}\right) = \frac{1}{4}\left(3e^{-3x} + e^{x}\right)$$

$$\int u_2' dx = \int \frac{e^{3x} - e^{-x}}{2} \times \frac{1}{-2} = \frac{-1}{4} \int e^{3x} - e^{-x} dx = \frac{-1}{4}\left[3e^{3x} + e^{-x}\right]$$

$$\Rightarrow u_2 = \frac{-1}{4}\left(3e^{3x} + e^{-x}\right)$$

$$\Rightarrow y_p = \frac{1}{4}\left(3e^{-3x} + e^{x}\right)(e^{x}) \cancel{+} \frac{-1}{4}\left(3e^{3x} + e^{-x}\right)(e^{-x})$$

$$y_p = \frac{1}{4}\left(3e^{-3x}e^{x} + e^{x}e^{x}\right) - \frac{1}{4}\left(3e^{3x}e^{-x} + e^{-x}e^{-x}\right)$$

$$y_p = \frac{3e^{-2x} + e^{2x}}{4} \cancel{+} - \frac{3e^{2x} + e^{-2x}}{4} = \frac{1}{4}\left[\left(3e^{-2x} + e^{2x}\right) \cancel{+} - \left(3e^{2x} + e^{-2x}\right)\right]$$

$$\Rightarrow y = C_1 e^{x} + C_2 e^{-x} + \frac{1}{4}\left[\left(3e^{-2x} + e^{2x}\right) - \left(3e^{2x} + e^{-2x}\right)\right]$$

Q9. $y'' - 9y = \frac{9x}{e^{3x}}$; $y_c = c_1 e^{-3x} + c_2 e^{3x}$

$m^2 - 9 = 0$

$\Rightarrow (m+3)(m-3) = 0$

$\Rightarrow y_1 = e^{-3x}$

$y_2 = e^{3x}$

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$m_1 = -3,\ m_2 = 3$

$$W_1 = \begin{vmatrix} 0 & e^{3x} \\ 9xe^{-3x} & 3e^{3x} \end{vmatrix} = -9xe^{-3x}e^{3x} = -9xe^{-3x+3x}$$
$$= -9xe^{0}$$
$$= -9x$$

$$W_2 = \begin{vmatrix} e^{-3x} & 0 \\ -3e^{-3x} & 9xe^{-3x} \end{vmatrix} = 9e^{-3x}e^{-3x}x = 9xe^{-3x-3x}$$
$$= 9xe^{-6x}$$

$$W = \begin{vmatrix} e^{-3x} & e^{3x} \\ -3e^{-3x} & 3e^{3x} \end{vmatrix} = 3e^{3x-3x} + 3e^{-3x+3x}$$

$W = 6$

$\Rightarrow \int u_1' dx = \int \frac{-9x}{6} dx = -\frac{3}{2}\int x\,dx = -\frac{3}{2}\left[\frac{x^2}{2}\right]$

$\Rightarrow u_1 = \frac{-3x^2}{4}$

$\Rightarrow \int u_2' dx = \int \frac{9xe^{-6x}}{6} dx = \frac{3}{2}\int xe^{-6x} dx$ } apply integration by part

$\Rightarrow \int uv\,dx = u\int v\,dx - \int\left[\frac{du}{dx} \times \int v\,dx\right]dx$

$= x\int e^{-6x}dx - \int\left[1 \times \int e^{-6x}dx\right]dx$

$= x\left(-\frac{1}{6}\right)e^{-6x} - \int -\frac{1}{6}e^{-6x} = -\frac{xe^{-6x}}{6} + \frac{1}{6}\left(-\frac{1}{6}\right)e^{-6x}$

$= -\frac{xe^{-6x}}{6} - \frac{e^{-6x}}{36} = \frac{-6xe^{-6x} - e^{-6x}}{36} = \frac{-e^{-6x}(6x+1)}{36}$

$\Rightarrow \int u_2' dx = \frac{3}{2}\left[\frac{-e^{-6x}(6x+1)}{36}\right] = \frac{1}{24}\left[-e^{-6x}(6x+1)\right] = u_2$

$\Rightarrow y_P = \left(-\frac{3x^2}{4}\right)\left(e^{-3x}\right) + \left(\frac{1}{24}\left[-e^{-6x}(6x+1)\right]\right)\left(e^{3x}\right)$

$y_P = \frac{-3x^2e^{-3x}}{4} + \frac{e^{3x}\left(-e^{-6x}(6x+1)\right)}{24} = \frac{-18x^2e^{-3x} + e^{3x}\left(-e^{-6x}(6x+1)\right)}{24}$

$y_P = -\frac{\left(18x^2e^{-3x} + e^{3x}\left(e^{6x}(6x+1)\right)\right)}{24}$

$$\Rightarrow y = C_1 e^{-3x} + C_2 e^{3x} - \frac{1}{24}\left[18x^2 e^{-3x} + e^{-3x}(6x+1)\right]$$

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Q16. $4y'' - y = e^{x/2} + 3$

$4m^2 - 1 = 0$;

$4m^2 = 1$

$m^2 = 1/4$

$\therefore m = \pm 1/2$

$\therefore y_c = c_1 e^{1/2 x} + c_2 e^{-1/2 x}$

$\Rightarrow y_1 = e^{1/2 x}$

$y_2 = e^{-1/2 x}$

For the particular solution;

$$W_1 = \begin{vmatrix} 0 & e^{1/2 x} \\ e^{x/2}+3 & \frac{1}{2}e^{1/2 x} \end{vmatrix} = e^{1/2 x}(e^{x/2}+3) = e^{x} + 3e^{1/2 x}$$

$$W_2 = \begin{vmatrix} e^{-1/2 x} & 0 \\ -1/2\, e^{-1/2 x} & e^{x/2}+3 \end{vmatrix} = e^{-1/2 x}(e^{x/2}+3) = 1 + 3e^{-1/2 x}$$

$$W = \begin{vmatrix} e^{-1/2 x} & e^{1/2 x} \\ -\frac{1}{2}e^{-1/2 x} & \frac{1}{2}e^{1/2 x} \end{vmatrix} = \frac{1}{2}e^{1/2 x - 1/2 x} + \frac{1}{2}e^{-1/2 x + 1/2}$$

$$= 1/2 + 1/2 = 1$$

$$\Rightarrow \int u_1' dx = \int e^{x} + 3e^{1/2 x}\, dx = e^{x} + 3(2)e^{1/2 x} = u_1$$

$$\Rightarrow \int u_2' dx = \int 1 + 3e^{-1/2 x}\, dx = x + 3(-2)e^{-1/2 x} = u_2$$

$\Rightarrow u_1 = e^{x} + 6e^{1/2 x}$

$\Rightarrow u_2 = x - 6e^{-1/2 x}$

$$\Rightarrow y_p = (e^{x} + 6e^{1/2 x})(e^{1/2 x}) + (x - 6e^{-1/2 x})(e^{-1/2 x})$$

$$y_p = e^{3/2 x} + 6e^{x} + xe^{-1/2 x} - 6e^{-x}$$

$$\Rightarrow y = c_1 e^{1/2 x} + c_2 e^{-1/2 x} + e^{3/2 x} + 6e^{x} + xe^{-1/2 x} - 6e^{-x}$$